

DD_FUL-09 - Equipment Swap Fulfillment - Developer Implementation Guide

Version: v1.0 **Bundle:** 06_fulfillment **Companion DD:** DD_FUL-09-Equipment_Swap_Fulfillment-v1.0

1. Goal

Implement the service/provisioning side of equipment swaps.

Keep the boundaries strict:

- OSR-RMA owns swap orchestration and recovery logistics.
- OSR-01/OSR-INSTANCE own stock and serial state.
- FUL-09 owns network/service provisioning for the swap.
- OSR-02 procurement is upstream replenishment only and is not called during a customer swap.

2. Runtime Flow

```
sequenceDiagram
    participant RMA as OSR-RMA-01
    participant FUL as FUL-09
    participant INST as OSR-INSTANCE-01
    participant PROV as PROV-INT/FUL-03
    participant APR as EM-CFG-04

    RMA->>FUL: POST /api/equipment-swap-fulfillments
    FUL->>INST: read source/target instances
    FUL->>FUL: Drools chooses steps
    FUL->>PROV: unbind old device
    FUL->>PROV: bind/provision new device
    FUL->>PROV: reconcile network state
    FUL-->>RMA: fulfillment outcome
    FUL->>APR: approval only for force-complete/override
```

3. Step 1 - Migrations

Create:

- equipment_swap_fulfillment
- equipment_swap_provisioning_step
- equipment_swap_fulfillment_reconciliation
- equipment_swap_exception

Add unique (operator_code, swap_request_id) so retries from OSR-RMA return the same fulfillment.

4. Step 2 - Start Fulfillment

Endpoint:

```
POST /api/equipment-swap-fulfillments
Idempotency-Key: esf-swap-001
Content-Type: application/json
```

Request:

```
{
  "operatorCode": "WIK",
  "swapRequestId": "swap-001",
  "woId": "wo-001",
  "subscriptionId": "SUB-700045",
  "accountId": "ACC-900045",
  "swapType": "DEFECT_SWAP",
```

```

    "technologyFamily": "GPON",
    "sourceInstanceId": "inst-old-001",
    "targetInstanceId": "inst-new-001",
    "homepassId": "HP-001"
}

```

Flow:

1. Validate request and idempotency key.
2. Load OSR-RMA swap request detail if needed.
3. Load source/target instances from OSR-INSTANCE.
4. Validate target instance is reserved/assigned by OSR flow.
5. Create equipment_swap_fulfillment in REQUESTED.
6. Emit EquipmentSwapFulfillmentRequested.

Do not call procurement. If stock is missing, OSR/WO should fail availability/reservation before this point.

5. Step 3 - Execute Fulfillment

Endpoint:

```
POST /api/equipment-swap-fulfillments/esf-001/execute
```

Flow:

1. Load fulfillment.
2. Build Drools fact.
3. Run equipment-swap-fulfillment-routing.drl.
4. Create provisioning step rows.
5. Execute steps in order through PROV-INT/FUL-03.
6. Reconcile.
7. If all pass, mark COMPLETED and callback OSR-RMA.
8. If failure, create exception and mark FAILED.

6. Step 4 - Drools Step Selection

Fact:

```

public record EquipmentSwapFulfillmentFact(
    String operatorCode,
    String swapType,
    String technologyFamily,
    String sourceSkuId,
    String targetSkuId,
    String homepassId,
    boolean sameTechnology,
    boolean relocationContext
) {}

```

Sample:

```

package com.sophix.fulfillment.equipment_swap

rule "GPON standard swap steps"
when
    $f : EquipmentSwapFulfillmentFact(technologyFamily == "GPON")
    $d : EquipmentSwapFulfillmentDecision()
then
    $d.addStep("GPON_UNBIND_OLD_ONT");
    $d.addStep("GPON_BIND_NEW_ONT");
    $d.addStep("GPON_SERVICE_RECONCILE");
end

rule "HFC modem swap preserves slot and port"
when
    $f : EquipmentSwapFulfillmentFact(technologyFamily == "HFC")

```

```

    $d : EquipmentSwapFulfillmentDecision()
  then
    $d.addStep("HFC_DISABLE_OLD_MAC");
    $d.addStep("HFC_AUTHORIZE_NEW_MAC_SAME_SLOT");
    $d.addStep("HFC_BROADHUB_RECONCILE");
  end

```

7. Step 5 - Provisioning Commands

For each step, call PROV-INT/FUL-03 adapter with deterministic idempotency key:

```

POST /api/provisioning/commands
Idempotency-Key: esf-001-GPON_BIND_NEW_ONT
Content-Type: application/json

{
  "operatorCode": "WIK",
  "commandType": "GPON_BIND_ONT",
  "subscriptionId": "SUB-700045",
  "homepassId": "HP-001",
  "equipmentInstanceId": "inst-new-001",
  "payload": {
    "serialNumber": "NEW123"
  },
  "originRefType": "EQUIPMENT_SWAP_FULFILLMENT",
  "originRefId": "esf-001"
}

```

Store command ref/status in equipment_swap_provisioning_step.

8. Step 6 - Reconciliation

After commands:

1. Query provisioning/network read adapter.
2. Compare expected target serial/MAC/entitlement/voice mapping with actual.
3. Store equipment_swap_fulfillment_reconciliation.
4. If any FAIL, create equipment_swap_exception.

Do not mark fulfillment complete until reconciliation passes or an approved force-complete occurs.

9. Step 7 - Callback OSR-RMA

```

POST /internal/swap-requests/swap-001/fulfillment-outcome
Content-Type: application/json

{
  "fulfillmentId": "esf-001",
  "outcome": "COMPLETED",
  "sourceInstanceId": "inst-old-001",
  "targetInstanceId": "inst-new-001",
  "completedAt": "2026-07-01T08:08:00Z"
}

```

OSR-RMA then continues its own lifecycle closure and stock/routing work. FUL-09 should not directly close OSR-RMA tables.

10. Step 8 - Exception Approval

For force-complete:

```

POST /api/approvals/evaluate
Content-Type: application/json

{
  "operatorCode": "WIK",
  "moduleCode": "FUL-09",
  "actionCode": "EQUIPMENT_SWAP_FORCE_COMPLETE",
  "subjectType": "EQUIPMENT_SWAP_EXCEPTION",
  "subjectId": "ese-001",

```

```

    "context": {
      "fulfillmentId": "esf-001",
      "exceptionType": "NETWORK_MISMATCH",
      "technologyFamily": "GPON"
    }
  }
}

```

If required:

1. call POST /api/approvals/requests;
2. store approval_request_id;
3. wait for callback;
4. apply approved override idempotently.

Camunda note: FUL-09 workflow may use Camunda for provisioning orchestration. EM-CFG-04 approval workflow is separate and only BPMN-backed when the approval policy says so.

11. Step 9 - Relocation/Shifting Context

During relocation:

1. SUB-WF-RELOCATION starts operation.
2. WO-01 SHIFTING handles field execution.
3. OSR/OSR-RMA handles old/new equipment movement and serial lifecycle.
4. FUL-09 provisions the target device/service binding.
5. Procurement is not called; missing stock blocks earlier OSR/WO availability.

This should be clear in code comments and API docs because otherwise teams may incorrectly create PO flows from customer jobs.

12. Worker Jobs

Job	Purpose
equipment-swap-fulfillment-execute	Executes pending fulfillment steps.
equipment-swap-reconcile	Retries reconciliation for pending/failed checks.
equipment-swap-exception-escalate	Notifies teams for unresolved exceptions.
equipment-swap-outbox-publish	Publishes FUL-09 events.

These run in the shared fulfillment/provisioning worker deployment.

13. Tests

- Start fulfillment idempotently from OSR-RMA.
- Reject if target instance is not reserved/assigned.
- GPON swap creates unbind/bind/reconcile steps.
- HFC swap creates old MAC disable/new MAC authorize/reconcile steps.
- Provisioning command retry uses same idempotency key.
- Reconciliation failure creates exception.
- Force-complete calls EM-CFG-04 when configured.
- Completed fulfillment calls OSR-RMA callback once.
- Relocation swap does not call OSR-02 procurement.